

INTERNATIONAL GCSE **BIOLOGY**

9201/2

Paper 2

Mark Scheme

June 2025

Version: 0.1 Pre-Standardisation



2 5 6 Y 9 2 0 1 / 2 / M S

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	mitochondrion		1	AO1 3.1.1a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.2	nucleus labelled correctly		1	AO1 3.1.1a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.3	to control the cell		1	AO1 3.1.1a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.4	<i>Measurement</i> 65 (mm)	allow conversion at any stage allow measurements in the range 64 – 66 mm	1	AO2 3.1.1a 6.3.1 6.3.3 6.3.10
	<i>Conversion</i> 65 000 (µm)		1	
	<i>Substitution</i> $\frac{65\,000}{50}$	allow $\frac{65}{0.05}$ allow correct substitution of an unconverted or incorrectly converted value	1	
	(×) 1300	allow a value correctly calculated using an unconverted or incorrectly converted value	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.5	B		1	AO1 3.2.4a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.6	to detoxify poisons		1	AO1 3.1.3c
	to make bile		1	3.4.3c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.7		ignore references to optimum temperature		AO2 3.2.4b
	(as temperature increases) (catalase activity) increases		1	
	then decreases		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.8	catalase / enzyme is denatured		1	AO3 3.2.4b
	(so) shape of <u>active site</u> has changed		1	
	or (so) catalase / enzyme can no longer bind to substrate			

Question	Answers		Extra information	Mark	AO / Spec. Ref.
01.9	Enzyme	Organ that produces the enzyme	Large molecule broken down	Small molecules produced	1 1 1 AO1 3.2.4c
	Amylase	salivary gland	starch	sugar / glucose	
	Lipase	small intestine	lipids	fatty acids and glycerol	
	Protease	stomach	protein	amino acids	

Total			17
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.1	leaf		1	AO1 3.1.4b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.2	mineral ions		1	AO1 3.1.4b 3.1.5g 3.2.2a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.3	stomata		1	AO1 3.2.2d 3.2.2e

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.4	same time		1	AO4 3.2.2c 3.2.2d
	same shoot	allow same type of plant allow same number of leaves allow (total) surface area of leaves	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.5	temperature		1	AO4 3.2.2c 3.2.2d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.6	increases		1	AO3 3.2.2c 3.2.2d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.7	evaporation		1	AO1 3.2.2d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.8	active transport requires energy	allow diffusion does not require energy	1	AO2 3.1.5a,b,f, g
	active transport is against a concentration gradient but diffusion is down the concentration gradient	allow description	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.9	take 3 / 4 plant shoots with different numbers of leaves	allow take 3 / 4 plant shoots with different sized leaves	1	AO4 3.2.2d
	measure the surface area of leaves	allow remove one leaf each time	1	
	measure loss of mass in same way		1	
	any one from: - control the temperature each time - repeat three times for each surface area and find a mean	allow description of controlling temperature	1	

Total			14
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.1	pancreas		1	AO1 3.4.5b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.2	blood glucose concentration increases		1	AO2 3.2.4a
	(because) sugar / glucose is absorbed from (small) intestine or (because) sugar / glucose is absorbed into the blood		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.3	insulin released (from pancreas)	ignore reference to glucagon and its effects	1	AO2 3.4.5a,b
	(so) glucose taken up into the cells		1	
	(then) glucose converted to glycogen (in muscle / liver cells)	allow excess glucose stored as lipid	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.4	water leaves the cells		1	AO2 3.1.5 d,e
	by osmosis	allow by diffusion through a partially / semi / selectively permeable membrane	1	
	(because) solution outside the cells is more concentrated than inside	allow the inside of the cells is more dilute than the outside	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.5	$\frac{1961}{3}$	allow 653.666667	1	AO2 3.4.5e 6.3.3 6.3.6
	$653.666667 \times 2 = 1307.333..$		1	
	1307		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.6	-6	allow -7 to -6	1	AO3 3.4.5e 6.3.11 6.3.12

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.7	any two from:		2	AO3 3.4.5e 6.3.12
	• Drug K reduces body weight over time • Drug K is more effective than the placebo or • placebo causes less weight loss than Drug K • Drug K has little / no effect on weight loss after 56 weeks • placebo has little effect on weight loss after 20 weeks • weight starts to increase slightly for placebo after 48 weeks correct use of supporting data from Figure 5 eg ‘Drug K decreased by a maximum of –16, but the placebo by –3.’ ‘after 64 weeks Drug K had reduced by –16 compared to –1.5 for the placebo’		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.8	Level 3: A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.		5–6	AO3 3.4.5e
	Level 2: Some logically linked reasons are given. There may also be a simple judgement.		3–4	
	Level 1: Relevant points are made. They are not logically linked.		1–2	
	No relevant content		0	
	Indicative content Support: • drug K significantly reduced body weight over the 64 week trial • large sample size of 1961 • patients randomly assigned drug K or placebo • patients taking drug K retain some loss in body weight after 128 weeks • patients on placebo return to starting body weight Limitations: ○ only tested over 64 weeks ○ only tested on patients over 18 years ○ could be the controlled diet causing change in body weight ○ could be the exercise causing change in body weight ○ those on placebo also lost some body weight ○ relying on patient to provide data and this might be unreliable ○ do not know long term effects ○ unlikely to get rid of obesity in the future as people would need to stay on the drug otherwise they will put the weight back on again For Level 3 , answers must consider support for the statement and limitations.			

Total			22
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.1	light energy is used for photosynthesis by water plant		1	AO1 3.3.1a 3.3.3e 3.2.1b
	(photosynthesis) produces glucose / sugar which is passed to water snail when plant is consumed / eaten	allow fats, lipids or proteins for glucose	1	AO2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.2	any one from: • not all parts of snail eaten • not all parts of snail digested • lost as faeces (from snail) • lost as urea / urine (from snail) • lost as carbon dioxide (from snail)	allow example such as shells not eaten	1	AO1 3.3.1b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.3	fish die as less / no oxygen available for respiration		1	AO2 3.3.4b
	(so) heron population decreases due to lack of food		1	AO3 3.3.4b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.4	any two from: • reproduces inside host cells to evade immune system • can infect multiple hosts to increase chance of reproduction / survival • millions of parasites produced to increase chance of (re)infection • parasites reproduce sexually increasing genetic variation (to survive changing conditions) • not easily killed by acid in stomach so reproduce in the gut • not killed by chlorine in drinking water so could infect human water supplies		2	AO3 3.3.2f

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.5	any one from: • more swimming in rivers • parasites survive longer in warm weather • more contaminated water consumed • parasites reproduce quicker in warm weather	allow other water sport suggestions	1	AO3 3.3.2f

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.6	any one from: • both are single-celled • both reproduce inside body cells • both reproduce sexually and asexually		1	AO3 3.3.2f

Total			9	
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Question	Answers	Extra information			Mark	AO / Spec. Ref.
05.1					1 1 1	AO2 3.3.2f
	Feature	Type of adaptation				
		Behavioural	Functional	Structural		
	Body shape			✓		
	Migrate thousands of kilometres to find beaches where eggs can be laid	✓				
	Maintain body temperature above cold water temperature		✓			

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.2	chemical	in either order	1	AO1 3.4.6e
	visual		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.3	any three from: - turtles migrate thousands of miles to find suitable / best sites to lay eggs - females hide eggs so that predators cannot (easily) find them - many baby turtles emerge at once to reduce the (overall) time for possible predation - innate behaviour of baby sea turtles to find ocean and (therefore) their habitat or innate behaviour of baby turtles to find ocean and avoid long exposure to sunlight / predators - baby turtles shelter in floating plants to reduce chance of being seen by predators or baby turtles shelter in floating plants to find food	if no other marks awarded allow large number of eggs to increase number of baby turtles produced	3	AO3 3.3.2f 3.4.6b,c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.4	Level 2: Scientifically relevant facts, events or processes are identified and given in detail to form an accurate account.		4–6	AO2 3.3.2a,c 3.3.3e 3.3.4d
	Level 1: Facts, events or processes are identified and simply stated but their relevance is not clear.		1–3	
	No relevant content		0	
	Indicative content - ice caps melt and sea levels rise - which submerge / floods beaches - destroys eggs in nests - reduces nest sites - seasons change and migration patterns affected - adults cannot find mates to reproduce - increased competition for mates - adults cannot find nesting beaches - increased competition for nest space - reduction in nests - warmer seas change distribution of food sources - reduction in food available for babies / adults - increased competition for food - warmer seas change distribution of predators - predators extend their range - new predators introduced to area (so) increased predation of adults / babies - warmer seas change distribution of plants - less shelter / food for baby turtles - increased predation of babies - fewer plants to absorb carbon dioxide in photosynthesis - warmer sand temperatures - change in ratios of males : females - potential loss of one sex - extreme weather damaging nests - eggs already in nest are damaged - eggs exposed to predators - reduction in number of babies For Level 2 answers must give both consequences of global warming and clear links to decline in sea turtle populations.			

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.5	as time / year increases the number of nests decreases	allow negative correlation	1	AO2 3.4.6b 6.3.12

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.6	suitable scale (must be more than half the space)		1	AO2 3.4.6b 6.3.9 6.3.11
	y-axis labelled		1	
	plots plotted	allow tolerance of +/- half a small square allow 4 or 5 correct plots for 1 mark	2	
	correct line of best fit	ignore line joined point to point with straight lines	1	

Total			20
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.1	any two from: • same size algal balls so the amount of algae is the same in each tube • test one tube at a time as any tube in front will block light and reduce light intensity • same volume of indicator so no effect on the colour change • use an LED lamp or heat shield to ensure temperature does not affect investigation • add a control tube to allow a comparison or as a base line • repeat each tube 3 times to identify anomalies • repeat at more distances to see the trend better • use a colorimeter / probe / sensor (to measure colour) so data is quantitative or not subjective • use a probe / sensor (to measure carbon dioxide) so colour of indicator does not affect results • add bungs to test tubes so carbon dioxide does not escape	allow description of a control tube	2	AO4 3.2.1 6.2.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.2	hot bulb can burn skin		1	AO4 3.2.1 6.2.2
	do not touch lamp or turn off lamp when not using		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.3	as light intensity increased from 40 cm to 10 cm (distance from the lamp) the indicator showed that more carbon dioxide was being used	allow converse argument for decreasing light intensity	1	AO3 3.2.1b 6.2.2
	photosynthesis uses the carbon dioxide (in the tubes) so the rate of photosynthesis must have increased (as the light intensity increased)		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.4	any one from:		1	AO4 3.2.1 6.2.2
	• use (different) coloured bulb(s) / LED(s) • use (different) coloured filter(s) in front of lamp • put (different) coloured transparent material(s) over lamp / beaker	ignore different coloured lamp allow named transparent material(s)		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.5	any one from: • count bubbles of oxygen in given time • measure volume of oxygen in given time • measure time taken for algal balls to float • use a probe / sensor to measure oxygen produced in given time	allow use a probe / sensor to measure carbon dioxide used in given time	1	AO4 3.2.1b 6.2.2

Total			8	
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